

Swastik Agarwala

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EDUCATION

Carnegie Mellon University

M.S. in Computer Science

Pittsburgh, PA

Expected Dec 2027

Purdue University

West Lafayette, IN

B.S. in Computer Science, Minors in Math & Economics (GPA: 3.90/4.0, Dean's List)

2022 – 2026

- **Coursework:** Algorithms, Statistical Machine Learning, Deep Learning, Artificial Intelligence, Software Engineering, Operating Systems, Object-Oriented Programming, Databases, Compilers
- **Competitive Programming:** ICPC-ECNA Regional; IOI Training Camp '21; Codeforces 1862 (Expert)

TECHNICAL SKILLS

Languages: Python, Go, Java, C++, C, TypeScript, JavaScript, Scala, SQL

Backend & Systems: FastAPI, Spring Boot, Express.js, Flask, REST APIs, Microservices, Multithreading

Web & Data: React, TanStack Query, PostgreSQL, TimescaleDB, Redis, MongoDB, Docker, AWS, Grafana, Git, Linux

AI/ML: LangGraph, LangChain, LLMs, RAG, vector search, NumPy

EXPERIENCE

Software Developer (Part-Time)

Jul 2024 – May 2026

Jyoti Broking Pvt. Ltd. (Proprietary Trading)

New Delhi, India

- Built and operated the proprietary trading desk's backend infrastructure as sole developer: 10+ interconnected Docker services spanning real-time data ingestion, computation, orchestration, and observability (Grafana LGTM)
- Developed real-time Go services ingesting high-throughput data feeds from the NSE, BSE, and MCX exchanges (~8 MB/s), with a begin/intraday/end-of-day orchestrator automating the full daily data pipeline
- Built a C++ service on a colocated exchange server that parses a binary market-data multicast feed and reconstructs the live order book in real time, with active-active redundancy and fault-tolerant recovery

AI Intern

May 2025 – Jul 2025

RMgX Technologies LLP

Remote

- Built a multi-agent travel assistant in LangGraph with coordinating sub-agents for recommendations, memory, and search, plus a multimodal RAG layer reasoning over PDFs, spreadsheets, and charts
- Cut token usage 30–40% via a dynamic context-caching memory system; built a recommendation engine using hybrid vector search (semantic + keyword) with metadata filtering

Software Engineering Intern

May 2024 – Jul 2024

SML Isuzu Ltd.

Chandigarh, India

- Shipped a React Native app for 400+ employees with QR scanning to digitize equipment-repair tracking; cut downtime 25% via a real-time notification system
- Architected a SAP integration to sync hardware specs, locations, and repair statuses to the mobile interface

Undergraduate Teaching Assistant

Jan 2024 – May 2026

Purdue University

West Lafayette, IN

- Mentored 60+ students across 3 courses in C/C++, algorithms, data structures, and systems programming; guided systems-level projects (custom shells, HTTP servers) and debugging with GDB

PROJECTS

MiniScala Compiler | *Scala, Compilers, Systems Programming*

Jan 2025 – May 2025

- Built a compiler and garbage collector translating MiniScala to assembly: lexical analysis, AST construction, type-checking, and comprehensive error handling
- Implemented an optimization pipeline with constant folding, dead-code elimination, and efficient value representations

Investify - Investing Learning Platform | *React, Spring Boot, PostgreSQL, AWS*

Aug 2024 – Dec 2024

- Built an investment-education platform with portfolio tracking, paper trading, and market analysis; led features in a 5-person team with code reviews and Git workflows
- Designed the React/TypeScript frontend with modular components and Hooks, using Redux Toolkit and RTK Query for state management and API caching